

1. Kinematics: Equations of Linear Motion

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The 4 Equations of Uniform Acceleration

1. $v = u + at$
2. $s = ut + \frac{1}{2}at^2$
3. $v^2 = u^2 + 2as$
4. $s = \left[\frac{(u + v)}{2}\right] \times t$

Where u = initial velocity (m/s), v = final velocity (m/s), a = acceleration (m/s²), t = time (s), s = displacement (m).

Motion Under Gravity

For bodies moving vertically upward: $a = -g$ (-9.8 m/s² or -10 m/s²).
At maximum height, vertical velocity $v = 0$ m/s.

2. Newton's Laws of Motion & Momentum

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Newton's Three Laws

First Law (Inertia): A body continues in its state of rest or uniform motion unless acted upon by an external net force.

Second Law: $F = dp/dt = ma$ (Force equals mass times acceleration).

Third Law: Action and reaction are equal in magnitude and opposite in direction.

Conservation of Linear Momentum

Total momentum before collision = Total momentum after collision:

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$